

- 15** The values of solubility products at 25 °C of three metal sulfides are shown in the table below. ΔH_{soln} of these metal sulfides is endothermic.

compound	value of K_{sp}
ZnS	1.1×10^{-21}
CuS	6.3×10^{-36}
Ag_2S	6.0×10^{-51}

Which statements are correct?

- 1 Addition of S^{2-} to the solution of Bi_2S_3 decreases the K_{sp} of Bi_2S_3 .
 - 2 Ag_2S is more soluble than CuS.
 - 3 Given a solution with $[\text{Cu}^{2+}] = [\text{Zn}^{2+}]$, CuS is precipitated out first on addition of S^{2-} .
- A** 1, 2 and 3
- B** 1 and 2 only
- C** 2 and 3 only
- D** 3 only